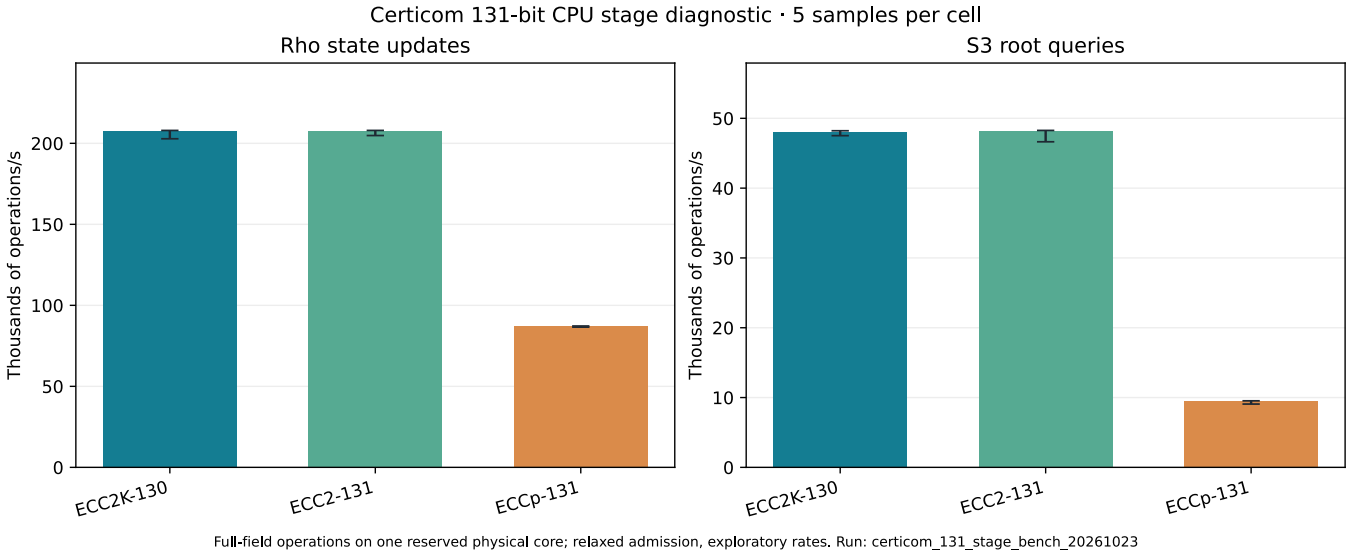
1. Certicom 131-bit rho and index-calculus stage benchmark

Run: 2026-10-23 UTC. Source base: `d7daaa6b42e7b5209e78c2407b2eaf35da25fbe4` plus `examples/certicom_131_stage_bench.rs` (SHA-256 `1e406f4cb146fa2739a5b74d93f151645414d46d5a7523bfac525f8c387f7acd`). Five samples per cell. CPU: Intel Xeon 6975P-C, sibling CPUs 14 and 30 reserved.



1.1. Observed CPU diagnostic

Rates are operations per second. Ranges are the minimum and maximum of five samples. Each rho sample used 200,000 state steps; each S3 sample used 50,000 quadratic-root queries in the exact 131-bit field.

Curve	Rho state steps/s	S3 root queries/s
ECC2K-130	207,822 (202,805–207,952)	48,080 (47,517–48,218)
ECC2-131	207,602 (204,808–207,962)	48,186 (46,639–48,266)
ECCp-131	87,055 (86,553–87,249)	9,522 (9,068–9,531)

The generic binary code has nearly equal stage rates on the two curves over the same field. The ECCp-131 generic CPU rho-step rate is 0.419 times ECC2K-130′s; its S3 query rate is 0.198 times ECC2K-130′s. These are stage ratios only.

1.2. Measurement and limits

The run took 58.14 seconds, of which 58.12 were user CPU seconds. Other processes used 28.69 CPU seconds across the machine (0.493 CPUs on average). A first run failed the default admission gate: 0.95 CPU seconds during 2 seconds against a 0.20 limit. The reported run used a 0.5-CPU diagnostic threshold and stayed just inside it. **The rates are exploratory, below the repository's strict isolation tier.** All five checksums per cell agree.

The rho unit includes partition selection, one group addition and two scalar coefficient updates for a planted Q = [73]G. The S3 unit includes coefficient construction and quadratic root solving. Distinguished-point storage, cycle escape, collisions, factor-base build, relation yield, linear algebra, descent and logarithm recovery were not measured. Curve generator/subgroup checks and positive S3 controls passed before timing.

1.3. GPU status and evidence

The planned ECC2K-130 GPU job never started: Spot launch returned `MaxSpotInstanceCountExceeded`, and on-demand launch was explicitly denied by the `SpotOnlyEC2Launch` policy. ECC2-131 and ECCp-131 have no matching GPU rho clients in this revision. No fresh GPU rate or full-size end-to-end IC result is reported.

Source records: `PROTOCOL.md`, `diagnostic-samples.csv`, `diagnostic-isolation.jsonl`, `admission.stderr`, `summary.tsv`, `summarize.rs`, and `plot.py` in `research/certicom_131_stage_bench_20261023/`. This report makes no fastest-implementation claim.